

Languages: English, Spanish,
Danish
Nationality: Cuban, Danish,
Norwegian

Einar Benjamin Jensen
MSc in Autonomous
Systems

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PROFILE

A hardworking and dedicated Mechatronics engineer with a strong passion for robotics, control systems, and automation technologies. Highly skilled in integrating electronics, programming, and mechanical design to develop intelligent robotic systems.

SKILLS

Programming	C, C++, Python, ROS2, Git, Object Oriented Programming
Robotics and Control	MATLAB/Simulink, Adaptive Nonlinear Control, System Identification, State-Space Modelling, Motion Planning, PyBullet
Machine Learning	Reinforcement Learning, PyTorch, TensorFlow, Gymnasium, Stable Baselines3
Embedded Systems & DSP	Embedded Programming, Linux/Ubuntu, Digital Signal Processing, Microcontroller Integration, Sensor interfacing, Sensor Fusion
CAD	Siemens NX, Creo Parametric, Autodesk Inventor, SolidWorks
Simulation	CFD Analysis in Star-CCM+, FEM Analysis in Ansys Workbench
Electrical	KiCad, LTSpice, PCB Prototyping, Analog Circuit Design
Soft Skills	Analytical problem-solving, strong collaborator, multidisciplinary teamwork

EDUCATION

Technical University of Denmark (DTU) MSc Honours Programme in Autonomous Systems	Sep 2026- Jan 2028
University of Southern Denmark (SDU) BSc Mechatronics Engineering	Sep 2022- Jan 2026 Honors 10.3/12 GPA
National University of Singapore (NUS) Exchange Programme, Robotics and Automation	Aug 2024-Dec 2024

EXPERIENCE

Bachelor Thesis Writer Novo Nordisk, Copenhagen, Denmark Thesis title: Modelling and Control of a Time-Pressure Membrane Valve System for Pharmaceutical Filling - Contributing to a novel filling line concept through a hybrid modelling approach combining physics and data-driven methods. - Designed an embedded system architecture for real-time sensor data acquisition for model validation - CFD simulation of the dispensing manifold to generate high fidelity data and support control strategy - Developed and validated nonlinear control strategies compared to baseline ON/OFF control.	Aug 2025-Jan 2026 Received Grade: 12/12
Manufacturing Automation Technology Intern Danfoss Drives, Gråsten, Denmark	Feb 2025-July 2025 Full-time

- Designed and deployed a robot assembly cell for human-robot collaboration complying with ISO standards
- Programmed serial-chain manipulators for kitting operations, involving inter-robot communication.
- Tested and redesigned frequency converter parts in CAD to optimise assembly efficiency per DFAA principles.
- Integrated and evaluated vision-based classification systems for component recognition, comparing commercial solutions with custom deep-learning models

PROJECTS

Reinforcement Learning for Serial-Chain Manipulator Control April 2026

- Built a custom OpenAI Gym environment for a 7-DOF Kuka iiwa arm in PyBullet
- Utilized a 17-dimensional observation space and a negative Euclidian distance reward function to guide policy learning
- Trained an on-policy PPO (Proximal Policy Optimization) agent using Stable Baselines3 with a custom MLP policy network [512,512,256,128] and ReLU activation achieving end effector positioning within 2cm of random fixed targets
- Two-stage training conducted using an additional reward term for angular velocity penalization to reduce joint oscillation near target

Deep Learning for Pose and Age Classification Apr 2025-Jun 2025

- Implemented an OpenPose-based python classifier with skeleton overlay for automated posture recognition in Jupyter Notebook
- Trained convolutional neural network for facial age classification using live video feed from external camera
- Integrated OpenCV for image processing and real-time tracking and display of predictions

Robot Manipulator Path Planning using STOMP Sep 2024-Nov 2024

- Developed a stochastic trajectory optimisation algorithm in MATLAB for collision-free motion planning of serial-chain manipulators
- Integrated forward kinematics using the Product of Exponentials formulation
- Maintained end-effector orientation constraints whilst simulating obstacle avoidance

Dynamic Modelling and Control of a 2-DOF Robotic Arm Sep 2024-Nov 2024

NUS Coursework

- Modelled a two-link robotic arm using Lagrangian dynamics and implemented the nonlinear system in MATLAB Simulink
- Designed and tuned a simple PID for trajectory tracking, evaluating its performance under varying inertia, Coriolis, and gravitational effects

Quadcopter Stabilisation Feb 2024-Jun 2024

Semester Project 4

- Modelled quadcopter dynamics in state-space representation
- LQI controller developed in MATLAB and Simulink and deployed in BeagleBone Blue
- Collected motor data on force-thrust characteristics for control system analysis and model validation
- Developed a Kalman filter for sensor fusion and state estimation, reducing process and measurement noise
- Designed and 3D printed a drone frame mechanically validated in Ansys

Autonomous Bartender AGV Sep 2023-Dec 2023

Semester Project 3

- Designed and built a tracked robot capable of delivering beverages to assigned seats along a bar table
 - Dual microcontroller setup (Arduino & ESP8266) for distributed control and real-time communication
 - Programmed PWM motor control and ultrasonic sensor-based obstacle avoidance in C++
 - Load cell integration for automatic initialization once drink is detected in holder
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